

Electric Chain Hoist

Technical data / extra information - generic portal datasheet

Document status

This document is a generic technical datasheet for portal reference. It does not replace the manufacturer manual, declaration of conformity, installation design, lift plan, risk assessment, inspection report or written scheme of examination.

1. Document control

Document title	Electric Chain Hoist Technical Datasheet	Portal filename	electric_chain_hoist_techni cal_datasheet.pdf
Equipment type	Powered lifting machine	Portal equipment ID	electric_chain_hoist
Prepared for	Stage Electrics compliance portal	Revision	Draft 1
Prepared by	Stage Electrics / portal author	Date	To be completed
Review status	For internal review and editing	Next review	To be completed

2. Equipment description

An electric chain hoist is a powered lifting appliance used to raise, lower or suspend loads such as truss, lighting bars, scenery or equipment. In theatre and event environments, it may be used as a temporary rigging hoist, a fixed installation hoist, or as part of a larger lifting system.

The hoist, hook, load chain, motor, brake, controls, suspension point, power supply and any lifting accessories form one complete lifting arrangement. The supporting structure and connection method are as important as the hoist itself.

Typical applications

- Lifting or suspending truss, lighting bars, scenery, cloths or production equipment.
- Temporary event rigging where hoists are installed, used and removed by competent riggers.
- Fixed theatre or venue installations where the hoist is part of a controlled lifting system.
- Multi-hoist lifting arrangements where load sharing, synchronisation and control arrangements must be considered.

3. Technical data fields

Complete this section with the actual manufacturer and site-specific information for the hoist being recorded.

Manufacturer		Model / type	
Serial number		Asset / ID number	
Safe Working Load / WLL		Duty rating / class	
Chain size / type		Number of falls	
Voltage / phase		Control voltage	
Control type		Speed	
Brake type		Limit switches fitted	Yes / No / N/A
Top suspension type		Bottom hook type	
Operating environment		Indoor / outdoor use	
Inspection interval		Thorough examination due	
User manual available	Yes / No	Declaration / certificate available	Yes / No
Location / project		Owner / responsible person	

4. Key components to identify

Component	Purpose	Information to confirm
Hoist body	Contains motor, gearbox, brake and load path components.	Manufacturer, model, capacity, condition and identification markings.
Load chain	Carries the suspended load through the hoist mechanism.	Correct chain type, condition, lubrication, twist, wear and compatibility.
Hooks and latches	Connect hoist to suspension point and load/accessory.	No distortion, cracks, wear, latch damage or incorrect seating.
Brake	Holds and controls the load when movement stops.	No drift, slipping, abnormal noise or delayed stopping.
Controls	Operate raising, lowering and emergency functions.	Correct direction, emergency stop, pendant/remote condition, labelling and safe response.
Limits	Restrict travel at upper and lower	Correct operation, correct setting and

	ends where fitted.	no reliance as a routine stop unless designed for that purpose.
Power/control cables	Supply power and control signals.	No cuts, crushed cable, strain, exposed conductors or unsuitable connectors.
Suspension point	Transfers the load into the supporting structure.	WLL/SWL, suitability, compatible accessories and condition of supporting structure.

5. Safe use and limitations

- Only use the hoist within its marked WLL/SWL and manufacturer instructions.
- Confirm the supporting structure, suspension point and lifting accessories are suitable for the total imposed load, including dynamic effects.
- Do not shock load, side load, drag loads, use damaged chain, or use the hoist to lift people unless it is specifically designed and certified for that use.
- Use only compatible chain containers, hooks, controls, power supplies and accessories approved for the hoist.
- Do not bypass, disable or rely incorrectly on limit switches, brakes, overload protection or control safety functions.
- Multi-hoist lifts require competent planning, suitable control arrangements and load sharing checks.
- Any modification, repair, change of use or unusual operating condition should be assessed by a competent person before use.

6. Inspection and maintenance guidance

Inspection principle

Inspection should combine pre-use checks, in-service inspections, maintenance checks and statutory thorough examination where required. The inspection regime should reflect manufacturer instructions, frequency of use, environment, duty cycle and risk.

Check level	Typical purpose	Examples of checks
Pre-use / visual check	Identify obvious defects before use.	Markings, hook condition, chain condition, controls, power cables, abnormal noise and correct operation.
In-service inspection	Detect deterioration during the service period.	Wear, corrosion, chain elongation, brakes, fasteners, controls, limits and records.
Planned maintenance	Keep the hoist in safe working order.	Lubrication, brake adjustment, electrical checks, chain inspection, manufacturer service tasks and replacement parts.
Thorough examination	Formal examination by a competent person where LOLER applies.	Condition, suitability, load path, safety-critical parts, defects, report,

		next due date and any limitations.
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7. What to check

- Check that the hoist is correctly rated for the load and that SWL/WLL markings are legible.
- Inspect the load chain for wear, twisting, corrosion, damaged links or contamination.
- Check hooks, latches, shackles and suspension fittings for distortion, cracks, wear or incorrect fitting.
- Check the body of the hoist for damage, missing covers, loose fixings or signs of impact.
- Test controls, emergency stop, direction buttons and limit functions where appropriate and safe to do so.
- Listen for abnormal noise, slipping, juddering or uneven movement during operation.
- Check that power, control cables and connectors are not damaged or strained.
- Confirm that the hoist has current inspection/thorough examination records and that defects have been actioned.

8. Defects requiring removal from service or competent review

- Missing or unreadable WLL/SWL, serial number or identification where traceability is required.
- Distorted, cracked, worn or opened hooks, damaged latches or unsafe hook seating.
- Twisted, elongated, corroded, kinked, contaminated or visibly damaged load chain.
- Brake slip, load drift, uncontrolled movement or inconsistent stopping.
- Damaged casing, missing covers, exposed electrical parts, damaged cable or unsafe connector.
- Limit switch failure, direction control fault, emergency stop failure or control labelling fault.
- Evidence of overload, shock loading, unauthorised repair, welding, modification or non-approved parts.
- No current inspection/thorough examination record where one is required, or unresolved reported defects.

9. Regulatory and good-practice context

This section summarises the key legal and good-practice context for a powered chain hoist used in theatre or production environments. It is intended as an aide-memoire and not a legal interpretation.

Source	Relevance to electric chain hoists	Portal note
LOLER 1998	Applies to lifting equipment and lifting operations at work. HSE states that lifting equipment should be fit for purpose, suitable, marked and, in many cases, thoroughly examined.	Relevant because the hoist is powered lifting equipment.
PUWER 1998	Applies to work equipment. HSE guidance highlights suitability, maintenance, inspection where deterioration could create risk, and adequate information and training.	Relevant because the hoist is work equipment used by employees/contractors.
BS EN 17206	Entertainment technology standard for machinery used on stages and production areas.	Relevant where the hoist forms part of entertainment machinery or a controlled lifting system.
ABTT Technical Standards	Recognised UK theatre industry good-practice guidance for places of entertainment.	Useful for venue-specific inspection, competence, maintenance and operating practice.

10. Site-specific notes

Installation / rigging notes	
Known limitations	
Required accessories	
Special inspection points	
Defects / actions	
Approved by / date	

11. Portal upload details

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12. References used for generic context

The following sources support the generic regulatory context. Site-specific use must still follow the manufacturer manual, competent-person advice and the agreed inspection regime.

Reference	URL
HSE - LOLER overview	https://www.hse.gov.uk/work-equipment-machinery/loler-overview.htm
HSE - Thorough examinations and inspections of lifting equipment	https://www.hse.gov.uk/work-equipment-machinery/thorough-examinations-lifting-equipment.htm
HSE - PUWER overview	https://www.hse.gov.uk/work-equipment-machinery/puwer-overview.htm
HSE - Inspection of work equipment	https://www.hse.gov.uk/work-equipment-machinery/inspection.htm